

34721 Linear Control Design 1, E23

Der anvendes en scoringsalgoritme, som er baseret på "One best answer"

Dette betyder følgende:

- Der er altid netop ét svar som er mere rigtigt end de andre
- Studerende kan kun vælge ét svar per spørgsmål
- Hvert rigtigt svar giver 1 point
- Hvert forkert svar giver 0 point (der benyttes IKKE negative point)

The following approach to scoring responses is implemented and is based on "One best answer"

- There is always only one correct answer – a response that is more correct than the rest
- Students are only able to select one answer per question
- Every correct answer corresponds to 1 point
- Every incorrect answer corresponds to 0 points (incorrect answers do not result in subtraction of points)

Page 1

Block diagram

A block diagram of a closed-loop system is shown in Figure 1.

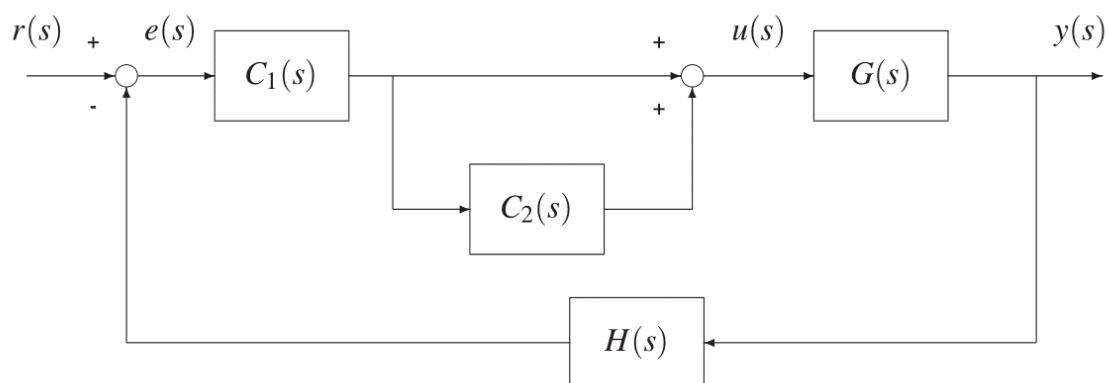


Figure 1.

$C_1(s)$, $C_2(s)$, $G(s)$, and $H(s)$ are dynamic systems. Further, it is assumed that the closed-loop system is stable.

Question 1

The open-loop transfer function $G_{open}(s)$ is given by:

☐ $G_{open}(s) = (C_1(s) + C_2(s))G(s)H(s)$

☐ $G_{open}(s) = \frac{C_1(s)}{1+C_2(s)}G(s)H(s)$

☒ $G_{open}(s) = C_1(s)(1 + C_2(s))G(s)H(s)$

☐ $G_{open}(s) = C_1(s)C_2(s)G(s)H(s)$

☐ $G_{open}(s) = C_1(s)C_2(s) + C_1(s)G(s)H(s)$

Question 2

The closed-loop transfer function $G_{cl}(s)$ from input r to output y is given by:

☐ $G_{cl}(s) = \frac{C_1(s)C_2(s)G(s)}{1+C_1(s)(1+C_2(s))G(s)H(s)}$

☐ $G_{cl}(s) = \frac{(C_1(s)+C_2(s))G(s)}{1+(C_1(s)+C_2(s))G(s)H(s)}$

☒ $G_{cl}(s) = \frac{C_1(s)(1+C_2(s))G(s)}{1+C_1(s)(1+C_2(s))G(s)H(s)}$

☐ $G_{cl}(s) = \frac{C_1(s)(1+C_2(s))G(s)}{1+C_1(s)C_2(s)G(s)H(s)}$

☐ $G_{cl}(s) = \frac{C_1(s)(1+C_2(s))G(s)H(s)}{1+C_1(s)(1+C_2(s))G(s)H(s)}$

Question 3

Let

$$C_1(s) = K_1$$

$$C_2(s) = \frac{K_2}{s}$$

$$G(s) = \frac{k}{\tau s + 1}$$

$$H(s) = \frac{1}{\tau_h s + 1}$$

What is the type and order of the open-loop system:

☐ Type 1 and order 2

☐ Type 0 and order 2

☐ Type 3 and order 1

☒ Type 1 and order 3

☐ Type 0 and order 3

Question 4

What is the static loop-gain K_0 for the open-loop system:

☐ $K_0 = K_1(1 + K_2)k$

☐ $K_0 = K_1(1 + K_2)$

☐ $K_0 = (K_1 + K_2)k$

☒ $K_0 = K_1 K_2 k$

☐ $K_0 = \frac{K_1(1+K_2)k}{\tau\tau_h}$

Question 5

$C_1(s)$ and $C_2(s)$ is the controller for the system. The controller is:

☒ PI controller

☐ P-lead controller

☐ P controller

☐ P-lag controller

☐ PI-lead controller

Dynamic system

Let a dynamic system be described by the following equations:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -ax_1 - bx_2 + cu$$

$$y = x_1$$

where u is the input to the system and y is the output from the system.

Question 6

The type and order of the system is:

☐ Type 1 and order 3

☐ Type 0 and order 3

☐ Type 1 and order 2

☒ Type 0 and order 2

☐ Type 2 and order 0

$$\dot{x}_1 = x_2 \Rightarrow \ddot{x}_1 = \dot{x}_2 = \ddot{x}_1 \quad \text{2nd-order}$$

Question 7

The open-loop transfer function $G(s)$ from u to y is given by:

☐ $G(s) = \frac{c}{s(bs+a)}$

☒ $G(s) = \frac{c}{s^2+bs+a}$

☐ $G(s) = \frac{c}{s(as+b)}$

☐ $G(s) = \frac{c}{as^2+bs+1}$

☐ $G(s) = \frac{1}{as^2+bs+c}$

Bode plot

A Bode plot for an open-loop stable system $G_{open}(s)$ be given shown in Figure 2.

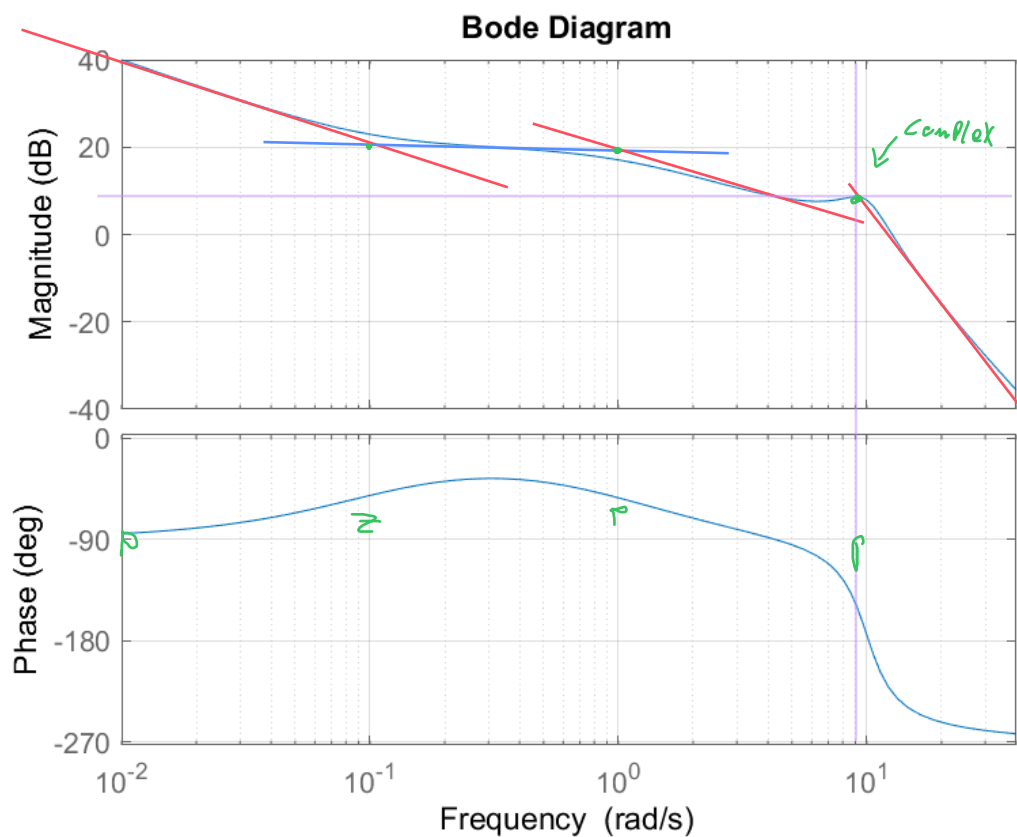


Figure 2.

-90° at origin

Question 8

The open-loop system $G_{open}(s)$ includes:

- ☐ Four complex poles.
 - ☒ One zero and four poles. One pole is in origo, one pole is real and two complex poles.
 - ☐ Three poles. One pole is in origo and two complex poles.
 - ☐ One zero and four complex poles.
 - ☐ One zero and four poles. One pole is in origo and the other poles are real.
-

Question 9.

What is the static loop-gain K_0 ?

- ☒ $K_0 = 1$

- ☐ $K_0 = 100$
 - ☐ $K_0 = 40$
 - ☐ $K_0 = 2.3$
 - ☒ $K_0 = \infty$
-

Question 10

What is the phase margin and gain margin:

- ☐ Phase margin $\approx 46^\circ$ and gain margin is infinity

- ☒ Phase margin $\approx -46^\circ$ and gain margin $\approx -7.6\text{dB}$

- ☐ Phase margin $\approx -46^\circ$ and gain margin $\approx 7.6\text{ dB}$
 - ☐ Phase margin is $\approx 90^\circ$ and the gain margin $\approx -7.6\text{ dB}$
 - ☐ Phase margin $\approx 46^\circ$ and gain margin $\approx 7.6\text{ dB}$
-

Question 11

The cross-over frequency ω_c and the π -frequency ω_π are:

- ☐ $\omega_c = 10.2 \text{ rad/sec}, \omega_\pi = 12.7 \text{ rad/sec}$
- ☐ $\omega_c = 0.35 \text{ rad/sec}, \omega_\pi = 10.2 \text{ rad/sec}$
- ☐ $\omega_c = 12.7 \text{ rad/sec}, \text{ and } \omega_\pi \text{ is not defined}$

☒ $\omega_c = 12.7 \text{ rad/sec}, \omega_\pi = 10.2 \text{ rad/sec}$

☐ $\omega_c = \omega_\pi = 10.2 \text{ rad/sec}$

Question 12

The system is now applied in closed-loop given by:

$$G_{cl}(s) = \frac{G_{open}(s)}{1+G_{open}(s)}$$

Let the output y be given by:

$$y = G_{cl}(s)r$$

where r is a unit step input.

We can say the following about the output y :

☒ $|y(t)| \rightarrow 1 \text{ as } t \rightarrow \infty$

☐ $|y(t)| \rightarrow \infty \text{ as } t \rightarrow \infty$

☐ $|y(t)| \rightarrow 0 \text{ as } t \rightarrow \infty$

☐ $|y(t)| \rightarrow k \sin(\omega t) \text{ as } t \rightarrow \infty, k \neq 0$

☐ $|y(t)| \rightarrow \text{constant} \neq 0 \text{ as } t \rightarrow \infty$

Question 13

A P-controller is now designed for $G_{open}(s)$ shown in Figure 2. The P-gain K_p is selected such that a phase margin at 90° is obtained.

K_p is:

☐ $K_p = 2.75$

☒ $K_p = 0.36$

☐ $K_p = 0.10$

☐ $K_p = 8.80$

☐ $K_p = 1.00$

A section of the Bode plot shown in Figure 2 is shown in Figure 3.

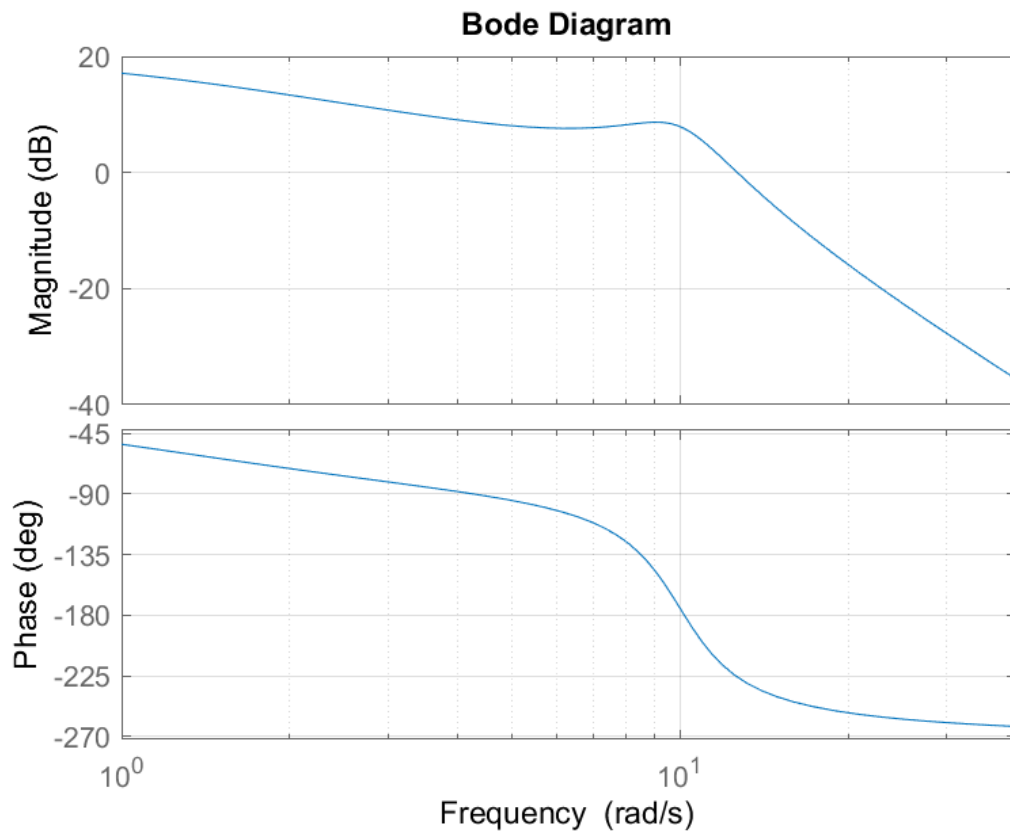


Figure 3.

A P-lead controller is designed for the system with $\alpha = 0.07$, and a phase margin at 60° . The cross-over frequency ω_c with this design is

- ☐ $\omega_c = 4.2 \text{ rad/sec}$
- ☐ $\omega_c = 7.6 \text{ rad/sec}$
- ☐ $\omega_c = 6.1 \text{ rad/sec}$
- ☐ $\omega_c = 12.6 \text{ rad/sec}$

☒ $\omega_c = 10.2 \text{ rad/sec}$

Question 15.

What is the main reason for using a P-lead controller:

- ☐ Will reduce the control signal
- ☐ Will always give a stable closed-loop system
- ☒ Will reduce the open-loop phase shift at low frequencies.
- ☐ Will increase the closed-loop bandwidth.
- ☐ Will increase the type of the system

Step response

Two questions about step responses.

Question 16.

Figure 4 shows a Bode plot of a stable system $G(s)$.

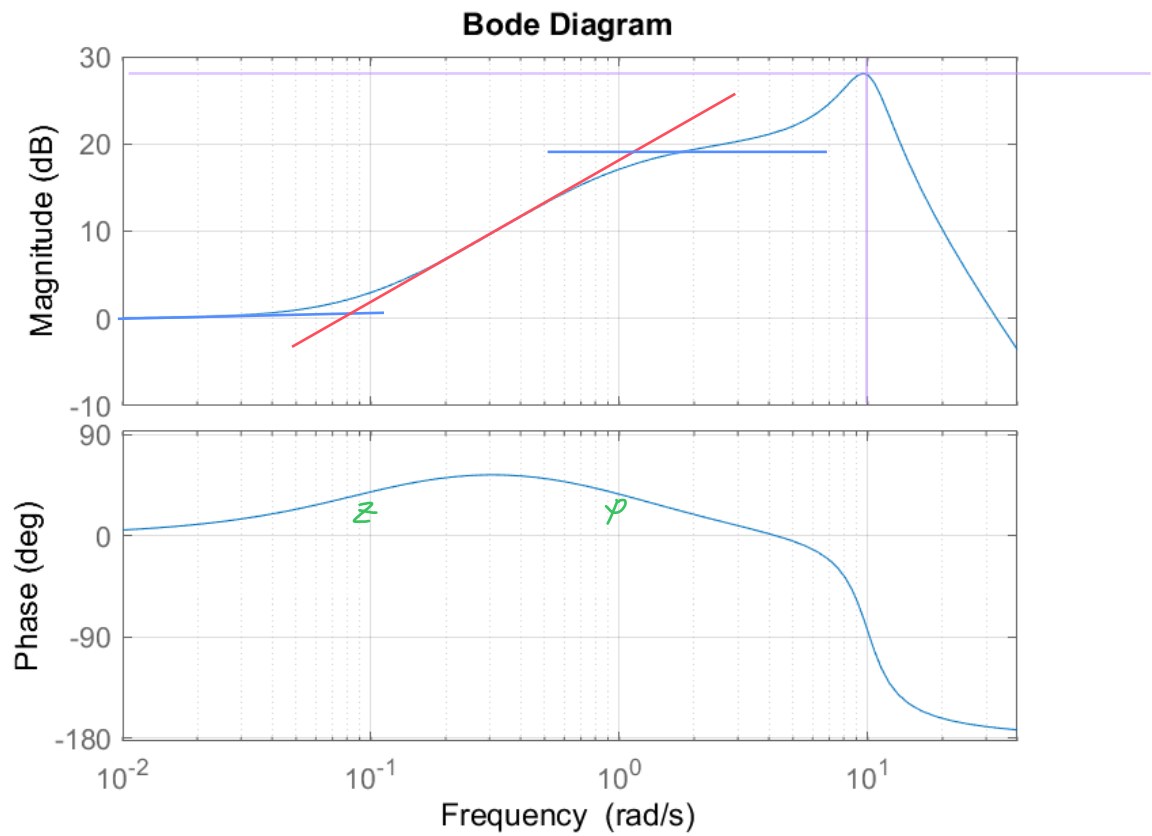
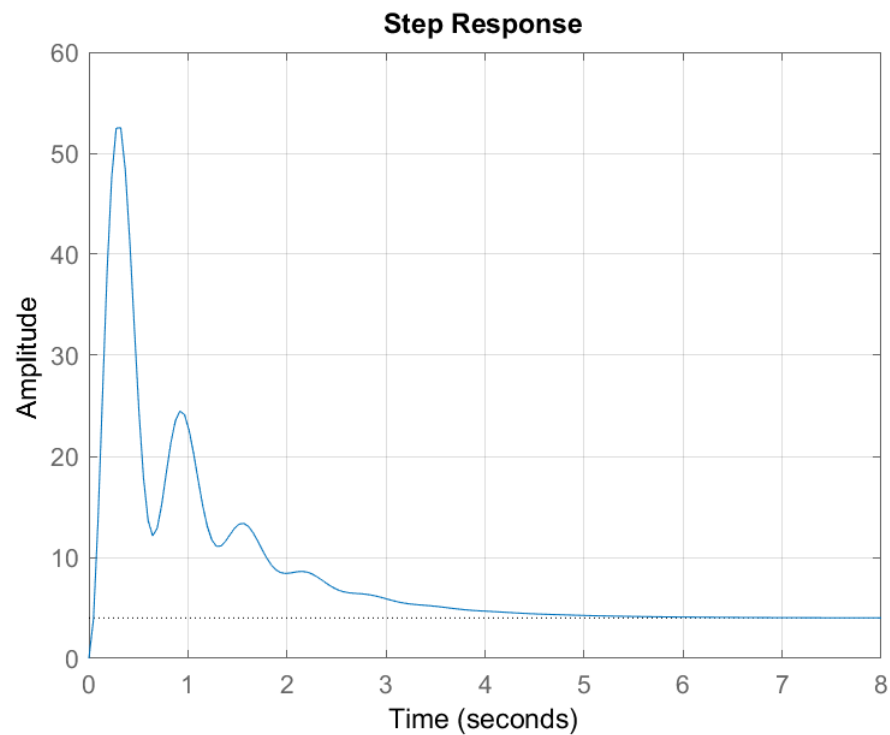


Figure 4.

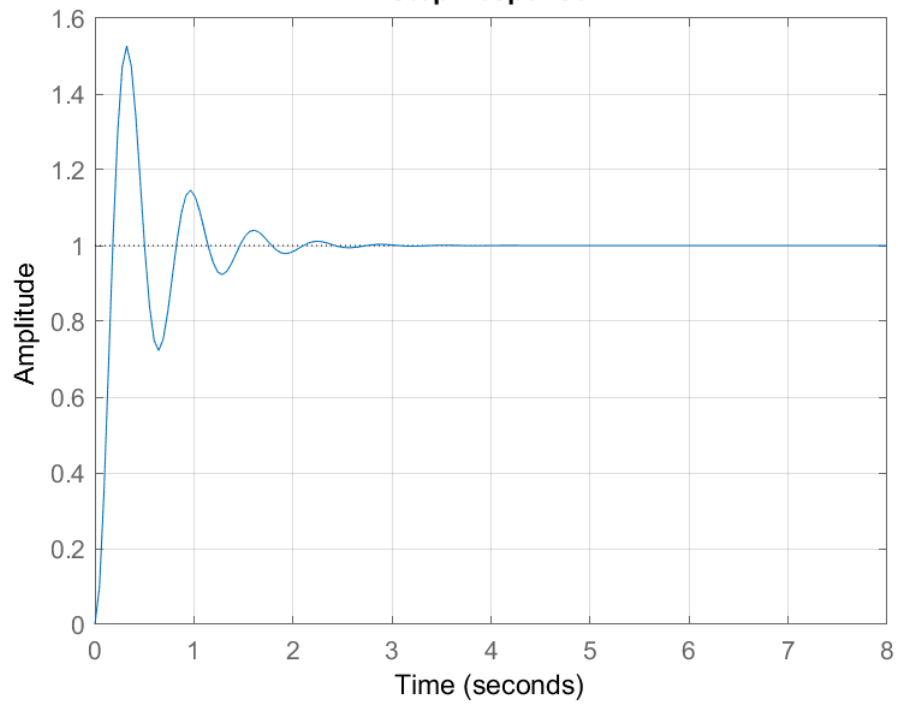
A unit step input is applied on $G(s)$. The response of the unit step input is:

○

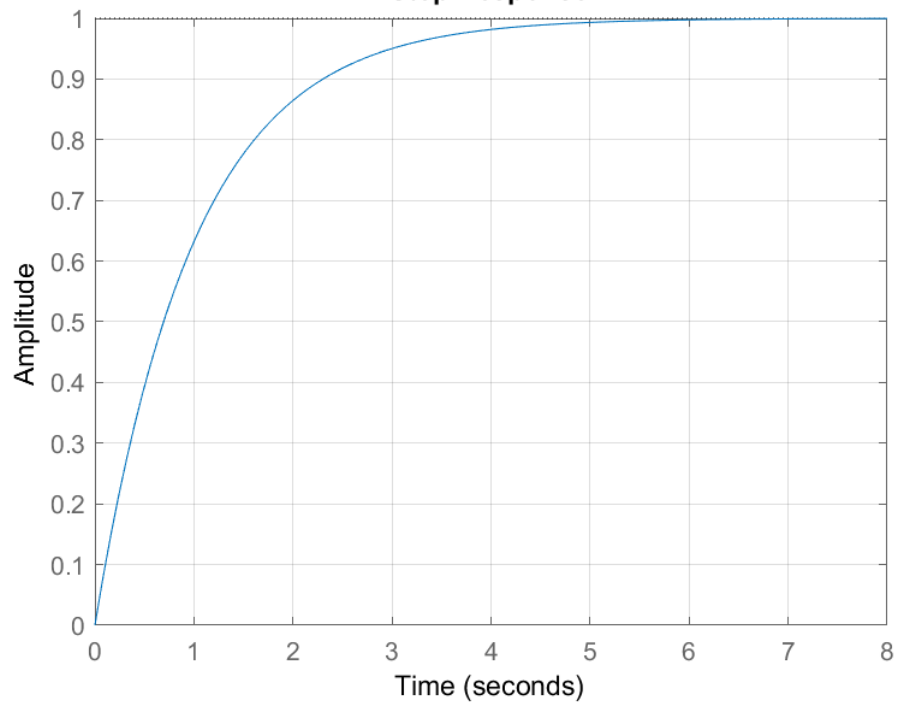


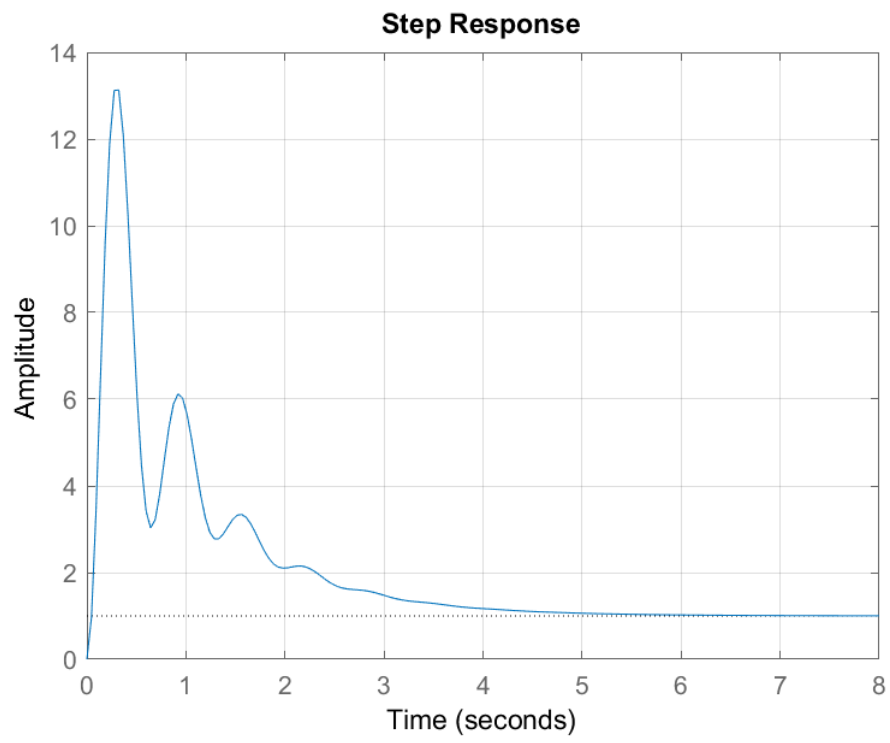
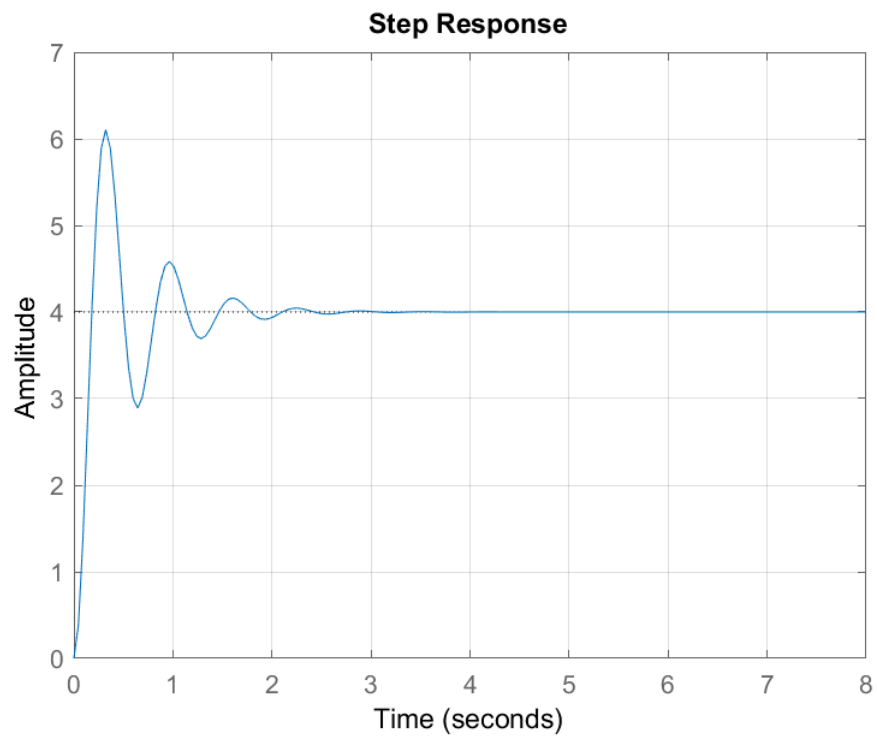


Step Response



Step Response





Question 17.

Consider the step response for a stable system shown in Figure 10.

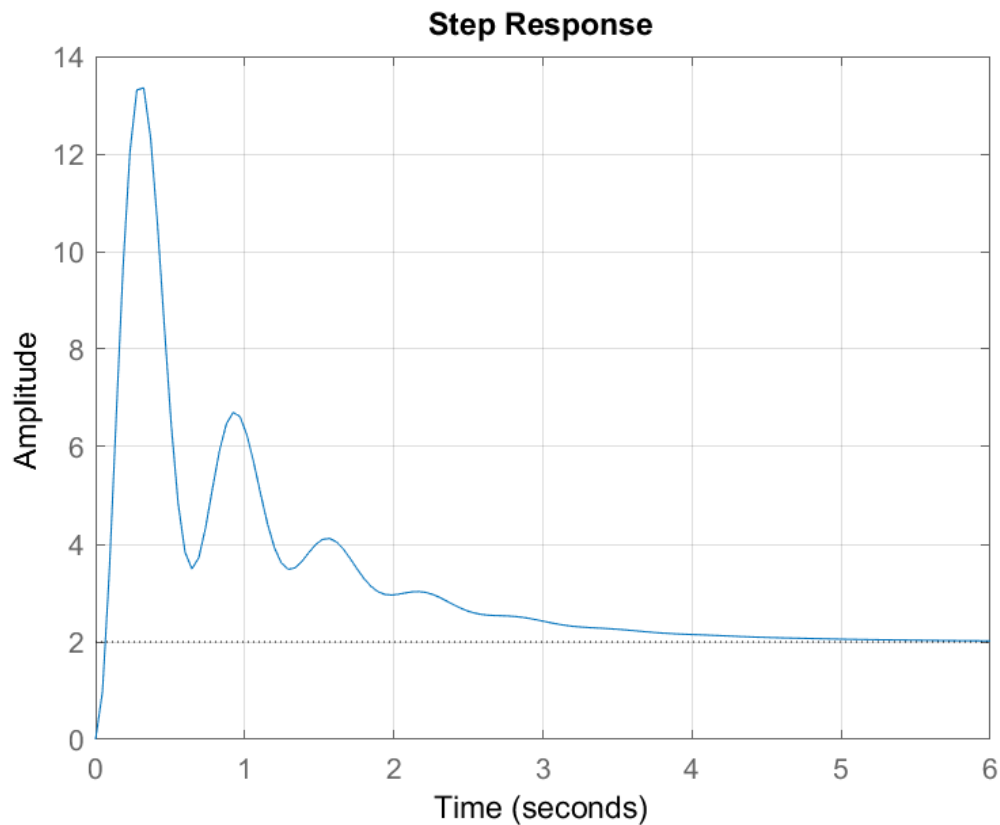


Figure 10.

The system includes a set of complex poles.

What is the damped eigenfrequency ω_d for the system?

☐ $\omega_d = 0.1 \text{ rad/sec}$

☐ $\omega_d = 0.6 \text{ rad/sec}$

☒ $\omega_d = 10 \text{ rad/sec}$

☐ $\omega_d = 3.14 \text{ rad/sec}$

☐ $\omega_d = 1.0 \text{ rad/sec}$

Steady-state error

Let a stable closed-loop system be given by the block diagram shown in Figure 11.

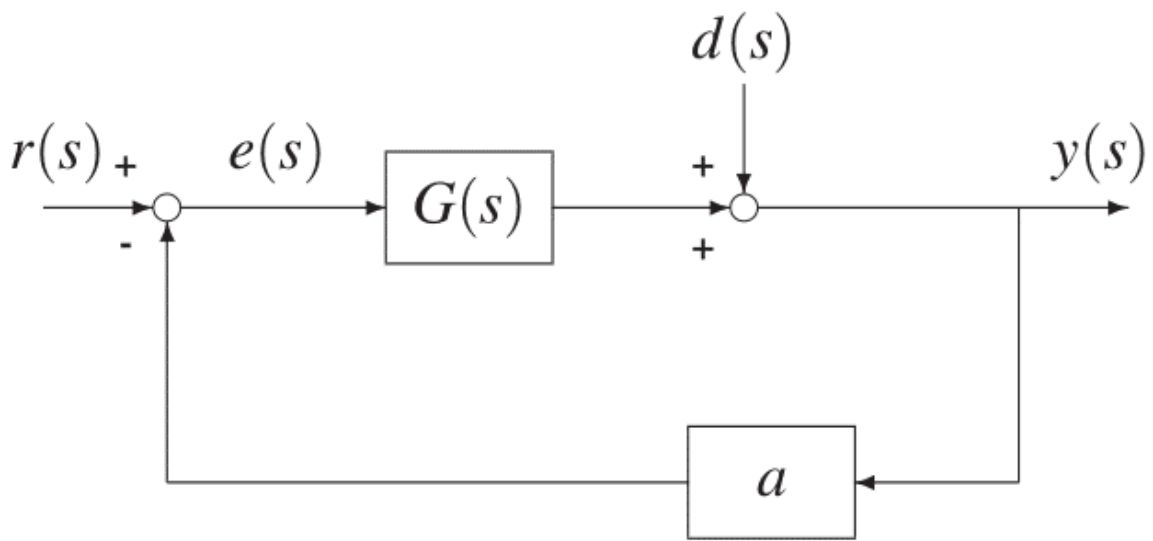


Figure 11.

$G(s)$ is a stable type 0 system with steady-state gain
 $G_{ss} = \lim_{s \rightarrow 0} G(s) = 1$

Question 18.

The steady-state error $e_{r,ss}$ for a unit step input at r is:

☐ $e_{r,ss} = \frac{1}{a}$

☒ $e_{r,ss} = \frac{1}{1+a}$

☐ $e_{r,ss} = 0$

☐ $e_{r,ss} = \frac{a}{1+a}$

☐ $e_{r,ss} = \frac{-1}{1+a}$

Question 19.

What is the steady-state value of the output y for a unit step input at d ?

☒ $y_{d,ss} = \frac{1}{1+a}$

$$\frac{1}{1+Ga}$$

☐ $y_{d,ss} = \frac{-1}{1+a}$

☐ $y_{d,ss} = \frac{1}{a}$

☐ $y_{d,ss} = 0$

☐ $y_{d,ss} = \frac{a}{1+a}$

Question 20.

The steady-state error $e_{d,ss}$ is $-\frac{2}{3}$ for a unit step input at d . What is a ?

☐ $a = \frac{5}{2}$

☐ $a = 5$

☐ $a = 1$

☒ $a = 2$

☐ $a = \frac{1}{2}$